Guided Lab 5.1 — README Instructions

📘 Course 02 – Introduction to Machine Learning for Cybersecurity

Module 05: Introduction to Neural Networks

🧠 Guided Lab 5.1

Building a Simple Neural Network Model

🎯 Lab Goal

In this guided lab, students will build and evaluate a simple neural network model using TensorFlow and Keras.

Students will:

- Load and prepare a dataset
- Build a simple neural network architecture
- Train the neural network model
- Generate predictions
- Evaluate model performance
- Interpret results in a cybersecurity or fraud detection context

This lab introduces the foundational workflow of deep learning using Python.

🚀 Note

The goal of this lab is not only to build a neural network model, but also to understand:

- How neural networks learn
- How predictions are generated
- How model performance is evaluated
- How AI supports cybersecurity systems

📁 Dataset

Use the provided dataset:

✅ `creditcard.csv`

Dataset Description

- `Class = 0` → Normal transaction
- `Class = 1` → Fraudulent transaction

This dataset is commonly used for fraud detection and anomaly detection tasks.

📌 Important Notes

- Update the dataset path to your local file location
- Ensure TensorFlow is installed correctly before starting
- Use comments throughout the notebook to explain your workflow
- Carefully interpret model evaluation metrics

🛠 Lab Activities

Part 1 — Import Libraries

Students will:

- Import TensorFlow and Keras
- Import Pandas, NumPy, and Scikit-learn
- Import visualization libraries

Part 2 — Load and Explore the Dataset

Students will:

- Load the dataset into Python
- Review dataset shape and columns
- Explore class distribution
- Check for missing values

Part 3 — Prepare the Data

Students will:

- Define features (`X`) and target (`y`)
- Split training and testing datasets
- Scale numerical features using `StandardScaler`

Part 4 — Build a Simple Neural Network

Students will:

- Create a sequential neural network model
- Add:
  - Input layer
  - Hidden layer(s)
  - Output layer
- Select activation functions

Example Concepts:

- Dense layers
- ReLU activation
- Sigmoid activation

Part 5 — Train the Model

Students will:

- Compile the neural network
- Train the model using training data
- Observe loss and accuracy during training

Important Concepts:

- Epochs
- Batch size
- Loss function
- Optimizer

Part 6 — Generate Predictions

Students will:

- Use the trained model to generate predictions
- Convert prediction probabilities into classification outputs

Part 7 — Evaluate Model Performance

Students will:

- Calculate:
  - Accuracy
  - Precision
  - Recall
  - F1 Score
- Generate a confusion matrix
- Interpret results

Part 8 — Interpret Results

Students should reflect on:

- Model strengths
- Model limitations
- False positives and false negatives
- Cybersecurity implications of predictions

📦 Submission Requirements

Submit:

✅ Completed Jupyter Notebook (`.ipynb`)

✅ All notebook cells executed with outputs visible

✅ Visualizations and evaluation metrics included

✅ Written interpretation questions completed

💡 Tips for Success

- Focus on understanding the neural network workflow
- Pay attention to how layers and activation functions work together
- Observe how training affects model performance
- Compare predictions to actual outcomes
- Remember that high accuracy alone may not be enough for imbalanced datasets

🔐 Cybersecurity Connection

Neural networks are increasingly used in:

- Fraud detection systems
- Intrusion detection systems
- Malware classification
- Threat intelligence analysis
- User behavior analytics

This lab demonstrates how deep learning models can help identify suspicious or fraudulent activity in cybersecurity environments.